

Brain alignment as a usable training signal: the internal extended manuscript

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Abstract

A reliable linear map predicts human language-network activity from a language model’s middle-layer activations, and this is almost always used to *measure* alignment after training. This thesis asks whether that alignment can instead be *used* as a training signal inside model compression, in the one cell the 2026 literature leaves empty: distilling a smaller student while brain alignment is in the objective, at a matched budget. An information budget argument predicts that a brain signal can act only as a weak prior, with whatever value it has concentrated in low-data and aggressively compressed regimes rather than acting as a second teacher. The program is a kill-gated ladder, and its first and foundational rung is established here: a trained model’s middle layer carries unique, out-of-sample variance about real language-network activity that a same-architecture untrained network does not, with a trained-minus-untrained gap of +0.0207 [95% CI +0.0205, +0.0209] for GPT-2 and +0.0277 [95% CI +0.0274, +0.0280] for Qwen2.5-0.5B at voxel scale, positive on 95 to 99% of reliable voxels after a capacity-fair low-level nuisance regression under contiguous splits [E006]. **[GAP: Abstract summary of the downstream findings (no preservation-by-default, the per-individual null at matched perplexity, the cross-subject-averaging confound and its confound-clean protocol, the quality law, the predictability ceiling, and the absence of a practical payoff) awaits the consolidation of R07–R14.]**

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Notation and terms

Encoding model

A ridge regression from a model’s per-stimulus features to measured brain activity, scored out-of-sample. The standard brain-alignment instrument.

Voxel / ROI

A voxel is one 3D measurement location in an fMRI scan; an ROI is a predefined region (here, the five left-hemisphere language regions of the ROI screen).

ΔR^2 (unique R^2)

The variance the language-model features add *on top of* the nuisance set: $R^2([Z_{\text{nuis}}, \text{LM}]) - R^2([Z_{\text{nuis}}])$. The semipartial R^2 , not the partial. The only predictive score this work reports.

Trained–untrained gap

The verdict statistic: the unique R^2 of a language-trained model minus that of a same-architecture untrained network, under identical nuisance and splits. Isolates what *language training* adds over random initialisation.

Z_{nuis} (nuisance set)

The low-level confounds subtracted in both arms of the gap: utterance length, position, word and phoneme rate, word duration, log word-frequency, and a static word-embedding block (eng1000). Surprisal is deliberately excluded (it is itself LM-derived).

Noise ceiling

The maximum variance any model could explain, set by the reliability of the neural measurement itself; reported here as split-half reliability of a held-out repeated story.

Permuted twin

The per-kind shuffled-target null that tests brain-*specificity*. It belongs to the lever question (whether the signal can be *moved*), not to the reality question, and so does not appear in the Results section written here.

$\mathbb{E}[Y \mid S]$

The stimulus-predictable mean response: the strong conditioning variable behind the ceiling question, as opposed to the weak low-level proxy Z_{nuis} .

$\mathcal{L}_{\text{brain}}$

The brain-alignment training loss: the term added to a language objective that rewards representations predictive of recorded brain activity.

1 Introduction

A decade of work has established a simple, reliable instrument: for a person and a language model reading the same text, a linear map predicts the person’s recorded brain activity from the model’s middle-layer activations, and how well it predicts on held-out data is the model’s brain-alignment score [1]. The instrument is almost always run in one direction: alignment is *measured*, treated as a property to observe after a model is trained. This thesis asks the inverted question: can that alignment be *used* as a training signal, a term added to a model’s objective that rewards representations the brain would predict, and does a model trained that way gain anything a model trained on text alone does not.

The inverted question only earns a thesis if it survives a hard prior: a language model already learns from on the order of 10^{12} tokens of dense, self-supervised supervision, while a language-fMRI dataset offers a few thousand noisy stimuli whose explainable variance is capped by a low noise ceiling. Training with an added loss term is, up to constants, a maximum-a-posteriori estimate in which the text enters as the likelihood and the brain term as a prior, not a second teacher. A prior worth, generously, 10^5 bits cannot install task knowledge against a likelihood worth 10^{12} ; what it can do is *select* among the many text-consistent solutions a large model admits, the way any weak-but-aimed regularizer does. Standard sample-complexity bounds then make a falsifiable prediction about where to look: if a brain prior helps at all, it helps most where the text signal is weakest (small models, scarce data, aggressive compression), and washes out where data is abundant, the benefit shrinking as the task data grows. This argument is developed formally, against the information-theoretic results it rests on, in the methods.

The bare inverted idea, using fMRI to shape a language model, is by 2026 already done, and saying so plainly is what locates the real contribution. It has been demonstrated for speech models [2], for text models [3], and causally: text models driven to be brain-*misaligned* at matched perplexity do substantially worse across a broad downstream suite, so alignment is load-bearing rather than incidental [4]. Every one of these interventions *grows* the model: it fine-tunes, adds adapters, or bolts on a brain-prediction head; none compresses one. The one study that compresses language models measures how quantization and pruning move an *observed* alignment score, and never puts alignment in the training objective, and never tests knowledge distillation at all [5]. Laid on two axes, what happens to the parameter budget (grow or shrink) and how alignment is treated (measured or optimized), the field fills three cells and leaves one empty: *shrink the model while alignment is in the objective*. That empty cell, alignment-guided distillation at a matched student budget, is where this thesis lives: the regime the weak-prior argument singles out as the place to test whether a brain prior is recoverable at all, not a place it is assumed to pay off.

The contribution depends entirely on the alignment score being a real signal rather than an artifact, because the field has repeatedly shown how easily it is not. Shuffled train-test splits leak through the temporal autocorrelation of fMRI, to the point that an *untrained* network can appear to “explain” brain data through sentence length and position alone [6]. So this work fixes its measurement standard before it measures anything: the reported quantity is never a raw score but the unique variance the model adds on top of a low-level nuisance set, on contiguous splits, with a same-architecture untrained network as the control [1, 7]. The program is then a kill-gated ladder, climbed only while each rung holds: first that the signal is real beyond confounds, then whether plain distillation preserves it, whether the loss is a usable lever, whether optimizing it yields a per-individual gain at matched perplexity, and whether any such gain buys a practical payoff.

This manuscript reports that ladder. Its one established positive, and the foundation everything else qualifies, is the first rung: a trained model’s middle layer carries unique, out-of-sample variance about real language-network activity that a same-architecture untrained network does not, and at voxel scale the effect is powered and unambiguous [E006]. The discipline that the

rest of the argument turns on is stated once and kept: *real* does not imply *movable*, and *movable* does not imply *useful*; this first result establishes only the first of the three. **[GAP: Introduction summary of the downstream findings (no preservation-by-default, the per-individual null, the averaging confound, the quality law, the predictability ceiling, no practical payoff) awaits the finding-reports R07–R14; do not state these verdicts here until their report layer exists.]**

2 Related work

The work on language models and the brain organizes cleanly on two axes, and the organization is what exposes the gap. The first axis is what an intervention does to the parameter budget: it either *grows* the model (fine-tunes, adds adapters, attaches a brain-prediction head) or *shrinks* it (quantizes, prunes, or distills into a smaller student). The second is how it treats brain alignment: either as a quantity *measured* after the fact, or as one *optimized* inside the training objective. Three of the four cells are occupied; the fourth is empty, and it is the thesis (Table 1).

Table 1: The field on two axes: what an intervention does to the parameter budget, and how it treats brain alignment. Three cells are occupied; the fourth, shrinking a model while alignment is in the objective, is empty in the 2026 literature and is where this work sits.

	Alignment measured	Alignment optimized
Model grown	classical brain-score [1, 8, 9]	brain-tuning [2–4]
Model shrunk	compression robustness [5]	this thesis (empty in the literature)

2.1 Grown models, alignment measured: the classical brain-score literature

The largest body of work fits encoding models to frozen, pretrained networks and reads off how alignment varies with architecture, scale, and training. Alignment is carried by structured middle-layer representations, with syntactic structure driving the layerwise trend and semantics more region-specific [1]. It rises with model scale and saturates in the few-billion-parameter range; at matched parameter count, scale rather than instruction-tuning moves it, which is why a base model is the appropriate alignment teacher [8]. Past that range the relationship can decouple and even reverse, as the largest models outgrow the human language network and track formal linguistic competence more than behavior [9]. This cell establishes that the signal exists and is non-trivial, but it never intervenes through alignment, so it cannot speak to whether alignment is usable.

2.2 Grown models, alignment optimized: brain-tuning, and why the bare inversion is scooped

Putting alignment in the objective of a model that is fine-tuned or grown has been done across modalities. For speech, a voxelwise loss fine-tunes compact encoders and improves both alignment and several downstream tasks, with a multi-participant variant reporting further gains at higher fMRI-data efficiency [2, 10]. For text, fMRI-augmented training of GPT-2 and LLaMA-2 with a cosine-similarity brain loss beats text-only baselines, though the reported gains are mixed across models and the splits are random rather than contiguous with no shuffled-brain control, so they would not pass the anti-confound bar adopted here [3]. Causally, driving text models *away* from alignment at matched perplexity damages performance across a 200-plus-task suite, and the same study’s positive arm shows that driving toward alignment helps, evidence that the recoverable capability is real [4]. The bare inverted idea is therefore not novel; what no study in this cell does is compress. Every one of them grows or holds the parameter budget, which leaves the compression question entirely open.

2.3 Shrunk models, alignment measured: the compression study that does not close the gap

One study compresses language models and tracks alignment, and it is the natural objection to the thesis, so it deserves a precise reading. It finds that alignment is preserved only for particular post-training quantizers on models of at least a few billion parameters; other quantizers, smaller models, and pruning all degrade it, and knowledge distillation is never tested [5]. Critically, this is *shrink* crossed with *measure*, not with *optimize*: alignment is observed under a fixed-function perturbation of the weights, never made a training target. Distillation is a different and lossier operation: it re-fits a smaller function from scratch against the teacher, the regime most likely to discard the middle-layer structure where alignment lives, so a result about quantization does not transfer to it. The data-processing inequality frames why the question is quantitative: a student can only lose alignment relative to its teacher, never create it, and in the narrow regime where preservation holds (particular quantizers on models of at least a few billion parameters) that bound is nearly tight; the open bet is that distillation, a lossier transform, sits well below tightness by default and that a brain objective pulls the student back toward the bound.

2.4 The empty cell, and the adjacent precedents that motivate it weakly

The remaining cell, shrink the model while alignment is optimized at a matched student budget, is unoccupied, and it is where this work sits. Two precedents from vision and audio are sometimes read as support for a brain regularizer in the low-data regime, but each weakens under scrutiny. A neural-data regularizer on a vision network improves accuracy and adversarial robustness, but a shuffled-label control reproduces the accuracy gain, so only the robustness benefit there requires real neural structure [11]. fMRI fine-tuning of a compact audio network improves a majority of downstream tasks, but the apparent low-data advantage carries no interaction test and no matched non-brain control, so it is suggestive rather than isolated [12]. These set the methodological warning the present work takes seriously, that the brain-specific increment must be measured against the control these studies skipped, more than they supply a transferable positive.

2.5 The measurement-validity bar, and the fMRI-free surrogate question

A parallel line argues that brain-alignment scores are easy to inflate and easy to over-read, and it sets the standard this work measures against. Shuffled splits leak through temporal autocorrelation, low-level and static features carry much of the apparent signal, and audited frozen-encoding scores can fail basic predictive and reliability gates [6, 13]. A related caution is that aggregate scores hide mechanism: the same headline number can be genuine semantics in one model and shallow phonology in another [14]. A more ambitious route would replace the live fMRI loss with a differentiable surrogate, but the literature does not yet agree on which way to push one: local intrinsic dimension correlates with alignment in opposite directions across language and vision studies, and a similarity-based surrogate is confounded by width and depth, with only neighborhood-overlap surviving null calibration [15–17]. A surrogate objective is therefore held as future work, since its very direction must first be validated against real fMRI.

3 Methods

3.1 The encoding model and the only score we trust

Alignment is measured with an encoding model: a ridge regression from a frozen model’s per-stimulus middle-layer activations X to recorded brain activity Y , fit on training stimuli and

scored on held-out ones. For activations $X \in \mathbb{R}^{n \times d}$ over n stimuli and a v -voxel response Y , the fitted map is

$$\hat{W} = \arg \min_W \|Y - XW\|_F^2 + \lambda \|W\|_F^2, \quad (1)$$

and the per-voxel score is the held-out correlation between predicted and measured response. A raw score of this kind is not reportable on its own, because a model can predict the brain for reasons that have nothing to do with language understanding: utterance length, word position, generic word identity. The quantity reported throughout is therefore the unique variance the model adds *on top of* a low-level nuisance set Z_{nuis} , the difference of two cross-validated fits,

$$\Delta R^2 = R^2([Z_{\text{nuis}}, \text{LM}]) - R^2([Z_{\text{nuis}}]), \quad (2)$$

a positive value of which says the contextual representation explains brain activity that length, rate, and static word identity cannot [E002].

3.2 What that quantity is, formally

In one sentence: the score is a lower bound on the information the model’s representation carries about brain activity beyond the low-level confounds. The unique-variance score is, formally, the linear-Gaussian estimator of a conditional mutual information. For a triple (X, Y, Z) , the conditional mutual information $I(X; Y | Z) = H(Y | Z) - H(Y | X, Z)$ is the dependence between X and Y that survives once the context Z is known [18]. It is the right object because plain mutual information conflates a genuine dependence with a spurious one in which X and Y are merely co-driven by a common source: when that holds, $X \perp Y | Z$ and the conditional term is zero even though the unconditional one is positive, so conditioning is what tells the two apart. The chain rule splits the question into a baseline and an increment, $I((X, Y); Z) = I(X; Z) + I(Y; Z | X)$, so the value of a learned representation Y over a cheap baseline X is the increment $I(Y; Z | X)$. Identifying the nuisance with X , the model’s middle layer with Y , and brain activity with Z , the reality claim is that $I(\text{LM}; B | Z_{\text{nuis}}) > 0$, which the unique-variance score estimates with ridge. For jointly Gaussian variables the conditional mutual information is a monotone function of the squared partial correlation, $I(X; Y | Z) = -\frac{1}{2} \ln(1 - \rho_{XY.Z}^2)$ [18, Ch. 8]. The reported score is the *semipartial* R^2 , the raw increment, which differs from the partial ρ^2 by a factor $1/(1 - R^2([Z]))$; both vanish together, so a positive semipartial score is equivalent to a positive conditional mutual information, which is the biconditional the argument needs. Because ridge is linear, the score captures the linearly decodable part of that dependence: a population-level lower bound on the true information, and the appropriate target for a thesis about the linear map.

The choice of conditioning variable scopes the whole program. Conditioning on the low-level nuisance Z_{nuis} establishes that the signal is real beyond low-level confound, not beyond the stimulus as a common cause; the stronger conditioning on the full stimulus-predictable response $\mathbb{E}[Y | S]$ is a separate, later question. The same chain rule with a stronger Z is what the downstream ceiling analysis evaluates.

3.3 Why brain data can only be a weak prior

The regime where a brain signal could matter follows from a budget argument, and it is worth stating because it predicts where the experiments should and should not expect an effect. Adding a brain term to a language objective is, up to constants, a maximum-a-posteriori estimate in which the text enters as the likelihood and the brain term as the log-prior. The likelihood’s pull scales with the supervision it carries, on the order of 10^{12} to 10^{13} bits across a training corpus, while the brain side offers 10^3 to 10^4 noisy stimuli whose per-voxel information is capped by a noise-ceiling channel capacity $I \leq \frac{1}{2} \log_2(1 + \text{SNR})$ with $\text{SNR} = r_{\text{max}}^2/(1 - r_{\text{max}}^2)$, putting the usable brain budget, as an order-of-magnitude estimate, around 10^4 to 10^6 bits

[18]. The data-processing inequality shrinks even that: the brain response is a noisy function of the same stimulus the model already trains on, so for the chain $S \rightarrow R(S) \rightarrow Y$ it carries no more about the optimal parameters than the stimulus does, $I(\theta^*; Y) \leq I(\theta^*; S)$ [18]. A prior this weak cannot teach new task structure against a likelihood this strong; what it can do is *select* among the high-dimensional manifold of near-equivalent text-optima that a large model admits, which is the precise sense in which a brain term is a regularizer rather than a teacher. A generalization bound makes the consequence quantitative: a prior concentrated on the right hypotheses tightens a sample-complexity bound by an amount that scales as the square root of the extra information it injects over the task data, and so vanishes as that data grows [19]. A weak-but-aimed brain prior is therefore predicted to help most in low-information regimes (small, under-trained, or aggressively compressed models) and to wash out where the text signal already dominates, which is the regime logic behind targeting distillation.

3.4 The verdict statistic: the trained–untrained gap

The quantity reported is not a trained model’s absolute unique-variance score but the *gap* between it and a same-architecture untrained network fit under identical nuisance and splits. The reason is that on naturalistic data an untrained network can score slightly positive through a shared-cause path: the stimulus drives both the response and the untrained features, leaving a residual dependence when the nuisance only partially captures the stimulus, so the absolute score is not a clean measure of what training adds. Both arms pass through the byte-identical nuisance subtraction, so the baseline term cancels in the difference and the gap directly isolates what *language training* contributes over random initialisation, controlling for architecture, tokenisation, and residual temporal structure [E006]. The gap is the statistic reported throughout.

3.5 Three controls, fixed as the standard

Three controls make that gap trustworthy, and they are applied uniformly. *Contiguous splits*: train and test are divided in blocks (block cross-validation on the ROI screen, whole-story-held-out cross-validation on the voxelwise run), because shuffling time-adjacent fMRI samples leaks near-duplicate timepoints across the split and inflates the score [E002][E006]. *A capacity-fair nuisance floor*: the wide contextual block and the wide static-embedding block are each reduced by PCA to the same rank, fit on the training fold only, so that when the model wins it wins on content rather than on carrying more dimensions; the few low-level scalars stay raw [E006]. *A held-out noise ceiling*: on the voxelwise run only reliable voxels are kept, and they are selected on a separate story repeated ten times and held entirely out of the cross-validation pool, scored by split-half reliability, so the estimate is not inflated by voxels that happen to correlate with the model by chance [E006]. Surprisal is deliberately left out of the nuisance set, since it is itself an LM-derived quantity and subtracting it would over-subtract the construct under test; log word-frequency, a static lexical scalar, is distinct and is subtracted [E006]. The permuted-target null, which tests brain-*specificity*, is not part of the reality question and enters only where the question is whether the signal can be moved.

3.6 Data and models

Two real datasets span the design. The ROI screen uses 1000 isolated sentences and the five left-hemisphere language regions of Tuckute et al. [20], a coarse but high-ceiling, train-participant-averaged benchmark whose isolated-sentence design sidesteps the temporal-autocorrelation leakage of naturalistic time series; its reported noise ceiling is $r \approx 0.35$ [E002]. The powered measurement uses subject UTS03 of the deep-sampled naturalistic story-listening dataset of LeBel et al. [21], with 20 stories, story-level folds, and 11,442 voxels retained at split-half reliability above

0.5 on a held-out repeated story [E006]. The models are GPT-2, GPT-2-medium, and Qwen2.5-0.5B, each read at its alignment-peak middle layer, with a same-architecture untrained network as the control in every comparison [E002][E006].

4 Results

4.1 The alignment signal is real, beyond confounds

A trained model’s middle layer carries unique, out-of-sample variance about real language-network activity that a same-architecture untrained network does not, and the effect survives the full low-level nuisance regression under contiguous splits. The ROI screen passes on every model tested, and the powered voxelwise run confirms it where the screen could only hint.

The ROI screen, on the five left-hemisphere language regions of Tuckute et al. [20] with 1000 isolated sentences over five contiguous folds, shows a positive trained-minus-untrained gap for all three models, with the untrained control negative at every layer and seed (Table 2). The signal is positive across a broad plateau of middle layers rather than at a single peak, and is absent from random features [E002].

Table 2: ROI screen on Tuckute 2024 (five LH language ROIs, five contiguous folds). Unique R^2 is the semipartial increment over the nuisance set; the gap is trained minus untrained. NC-norm is the gap as a fraction of the $r \approx 0.353$ noise ceiling.

Model	Best layer	Trained ΔR^2 (mean $\pm$ SD)	Untrained ΔR^2	Gap	NC-norm	Source
GPT-2 (12L)	L7	+0.020 $\pm$ 0.008	−0.011	+0.030	0.056	[E002]
GPT-2-medium (24L)	L14	+0.019 $\pm$ 0.005	−0.014	+0.033	0.053	[E002]
Qwen2.5-0.5B (24L)	L12	+0.036 $\pm$ 0.010	−0.014	+0.050	0.102	[E002]

The powered voxelwise run, on subject UTS03 of LeBel et al. [21] over 20 naturalistic stories and 11,442 voxels selected at split-half reliability above 0.5 on a held-out repeated story, gives a trained-minus-untrained gap of +0.0207 [95% CI +0.0205, +0.0209] for GPT-2 and +0.0277 [95% CI +0.0274, +0.0280] for Qwen2.5-0.5B, with 95% and 99% of reliable voxels positive respectively and the untrained unique R^2 at −0.017 in both (Table 3) [E006]. The confidence intervals exclude zero, and the gap is carried by language training rather than by architecture or the temporal structure the nuisance floor absorbs [E006].

Table 3: Powered voxelwise run on LeBel UTS03 (20 stories, story-level folds). The gap is the trained-minus-untrained unique R^2 ; CIs are bootstrap.

Model	Layer	Trained – untrained gap [95% CI]	% reliable voxels gap > 0	Source
GPT-2	L7	+0.0207 [95% CI +0.0205, +0.0209]	95%	[E006]
Qwen2.5-0.5B	L12	+0.0277 [95% CI +0.0274, +0.0280]	99%	[E006]

Three controls keep the positive honest, and the screen and the powered run are complementary on them. The ROI screen uses isolated sentences, which removes the temporal-autocorrelation inflation that contaminates naturalistic time series, so its positive is harder to obtain rather than easier [E002]. The voxelwise run reintroduces naturalistic stimuli but adds whole-story-held-out cross-validation, so test timepoints are never adjacent to training timepoints, and voxel selection on a separate held-out repeated story, so the score cannot be inflated by voxels that correlate with the model by chance [E006]. The nuisance set subtracted in both arms is the low-level set most likely to manufacture a false positive: word and phoneme rate, word duration and length, log word-frequency, and a static word-embedding block [E006]. Surprisal and imageability are deliberately left unsubtracted, so a residual contextual nuisance is

not ruled out by the floor alone; what rules it out is the same-architecture untrained network, which sees the identical set and scores negative at -0.017 , so architecture, rate, and residual temporal structure cannot manufacture the gap [E006].

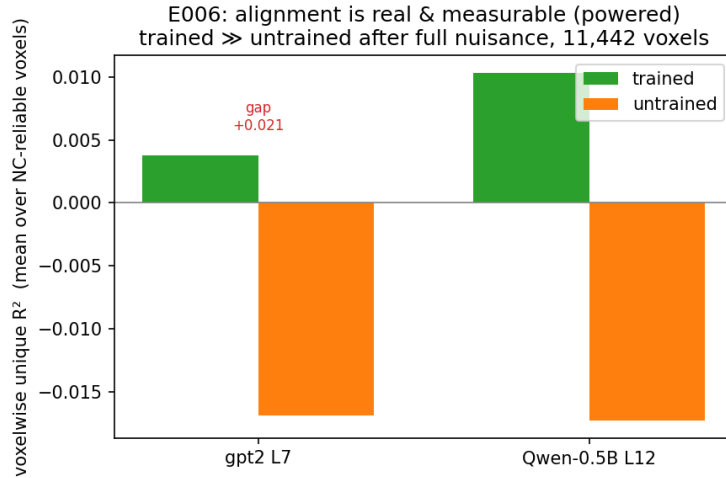


Figure 1: The powered reality result: the trained-minus-untrained unique- R^2 gap is positive across the overwhelming majority of reliable voxels on LeBel UTS03 [E006].

The signal is real beyond confounds on real neural data, with the breadth carried by the three-model ROI screen and the power carried by the deeply-sampled single subject UTS03 at voxel scale; the cross-subject generality of the powered effect is a question the single-subject design leaves open, not one it settles. Qwen aligns more strongly than either GPT-2 variant, while the two GPT-2 variants are nearly tied despite a threefold parameter difference, so this single cross-family contrast does not by itself establish a quality ordering; that question is taken up later [E002]. This is the one load-bearing positive of the thesis, and it licenses the questions the rest of the program asks without pre-judging them: whether the signal can be moved, and whether moving it buys anything. The discipline the whole arc turns on, stated once: real does not imply movable, and movable does not imply useful, and this result establishes only that the signal is real.

4.2 Plain distillation does not preserve alignment for free

[GAP: Results Q1 awaits finding-report R07 (source E003): the monotone alignment gradient, the quality co-variation, and the unresolved KD-specific shedding. Do not consolidate from the experiment record before R07 is written.]

4.3 The brain loss is a lever, but a weak and quality-confounded one

[GAP: Results Q2 awaits finding-report R08 (source E004): the brain-specific lever on the strong aligner and its fragility.]

4.4 No per-individual gain at matched perplexity, and the averaging confound

[GAP: Results Q3 awaits finding-reports R09–R13 (sources E005, E008, E011, E013, E017, E019, E020): the per-individual null, the cross-subject-averaging confound, the method-general failure, the quality law, the stimulus-predictability ceiling, and the external reproduction. The parked draft reports/_pending-Q3 reclaims R09.]

4.5 No practical payoff

[GAP: Results Q4/A3 awaits finding-report R14 (source E009): the bounded OOD-perplexity null and its null-by-construction reading.]

5 Discussion

[GAP: Discussion awaits the consolidated Results arc (R07–R14). The intended spine: the positive contribution is a measurement-validity finding plus a confound-clean protocol, not a bare null; the cross-subject-averaging confound manufactures apparent brain-specificity, and the matched-perplexity, per-kind permuted-twin, per-subject protocol is the control the alignment-training literature lacks. Write only once the Results sections it interprets exist.]

6 Limitations

[GAP: Limitations awaits the consolidated Results arc. The intended content, each tied to its result: the absolute unique variance is a few percent of the noise ceiling (a scale fact, not a weakness); the reality claim is "beyond the nuisance actually subtracted," with surprisal held out by design; the per-individual null is measured at one power level on one cohort; the stimulus-predictability ceiling is instrument-limited rather than closed. Write once the results exist.]

7 Conclusion

[GAP: Conclusion awaits the consolidated Results arc. It should restate the established positive (the alignment signal is real and powered beyond confounds) and the program's net finding once R07–R14 are consolidated, without reaching past the report layer.]

Checkpoint log

2026-06-16: Checkpoint 1, the manuscript is opened

This is the first checkpoint: the extended manuscript is created and the layers that already trace to a written finding-report are drafted. The science behind it was produced over the preceding working sessions and recorded in the experiment files; this checkpoint consolidates the part of that record that the report layer has so far narrated.

What is folded in at this checkpoint. The measurement apparatus and the one established positive result: a trained language model's middle layer carries unique variance about real language-network activity beyond a low-level nuisance regression, and a same-architecture untrained network does not [E006]. This is consolidated from the finding-report on the reality of the signal (R06), which itself rests on the ROI screen [E002] and the powered voxelwise run [E006]. The front matter (Introduction, Related work) is consolidated from the background reports: the first-principles case for the brain as a weak prior (R03), the gap analysis that locates the contribution in the empty *shrink* $\times$ *optimize* cell (R04), and the synthesis of the measurement literature (R01).

What is deliberately left open. The Results sections beyond the reality of the signal (whether plain distillation preserves alignment, whether the loss is a usable lever, the per-individual null, the quality law, the predictability ceiling, and the practical-payoff question) are marked with \gap because their finding-reports (R07–R14) are not yet written. Knowledge flows down only:

a number is narrated in a report before it is consolidated here, so these sections wait on the report layer rather than reaching past it into the raw experiment records. The abstract is provisional for the same reason: it can state the question and the reality result, but not yet summarise an arc whose later findings have no narration layer.

How this checkpoint was built. The draft was written through the scientific-writing skill on the full loop, with the review pass run continuously: adversarial critique agents argued against the reasoning and the prose quality section by section, and the objections that held were addressed before the section was kept.

References

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